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The Golden Age

in which black holes are found
to spin and pulsate,
store energy and release it,
and have no hair

The year was 1975; the place, the University of Chicago on the south side of the city, near the shore of Lake Michigan. There, in a corner office overlooking 56th Street, Subrahmanyan Chandrasekhar was immersed in developing a full mathematical description of black holes. The black holes he was analyzing were radically different beasts from those of the early 1960s, when physicists had begun to embrace the concept of a black hole. The intervening decade had been a *golden age* of black-hole research, an era that revolutionized our understanding of general relativity's predictions.

In 1964, at the beginning of the golden age, black holes were thought to be just what their name suggests: holes in space, down which things can fall, out of which nothing can emerge. But during the golden age, one calculation after another, by more than a hundred physicists using Einstein's general relativity equations, had changed that picture. Now, as Chandrasekhar sat in his Chicago office, calculating, black holes were regarded not as mere quiescent holes in space, but rather as dynamical objects: A black hole should be able to spin, and as it spins it should create a tornado-like swirling motion in the curved spacetime around itself. Stored in that swirl should be enormous energies, energies that nature might tap and use to power cosmic explosions. When stars or planets or smaller holes fall into a big hole, they should set the big hole pulsating. The horizon of